

## Ideation Phase

### Define the Problem Statements

|               |                                                                   |
|---------------|-------------------------------------------------------------------|
| Date          | 24 June 2026                                                      |
| Team ID       | SWTID-2026-4219                                                   |
| Project Name  | Visualization Tool for Electric Vehicle Charge and Range Analysis |
| Maximum Marks | 2 Marks                                                           |

#### Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

|                            |                                                                                                |                                                                             |
|----------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>I am</b>                | Describe customer with 3-4 key characteristics - who are they?                                 | Describe the customer and their attributes here                             |
| <b>I'm trying to</b>       | List their outcome or "job" the care about - what are they trying to achieve?                  | List the thing they are trying to achieve here                              |
| <b>but</b>                 | Describe what problems or barriers stand in the way - what bothers them most?                  | Describe the problems or barriers that get in the way here                  |
| <b>because</b>             | Enter the "root cause" of why the problem or barrier exists - what needs to be solved?         | Describe the reason the problems or barriers exist                          |
| <b>which makes me feel</b> | Describe the emotions from the customer's point of view - how does it impact them emotionally? | Describe the emotions the result from experiencing the problems or barriers |

Reference: <https://miro.com/templates/customer-problem-statement/>

#### Example:

|            |                          |                      |                                                                 |                     |
|------------|--------------------------|----------------------|-----------------------------------------------------------------|---------------------|
| I am       | I'm trying to            | But                  | Because                                                         | Which makes me feel |
| a traveler | book flights on my phone | it takes a long time | The website is not responsive and doesn't have a mobile version | Frustrated          |

| <b>Problem Statement (PS)</b> | <b>I am (Customer)</b>                                                                              | <b>I'm trying to</b>                                                                                                      | <b>But</b>                                                                                     | <b>Because</b>                                                                                                        | <b>Which makes me feel</b>                                                                                            |
|-------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| PS-1                          | A potential Electric Vehicle (EV) buyer who wants to compare different EV models before purchasing. | Compare electric vehicle range, charging speed, battery efficiency, price, and brand performance in one place.            | The information is scattered across multiple websites and is difficult to compare effectively. | There is no single interactive platform that presents all important EV information through meaningful visualizations. | There is no single interactive platform that presents all important EV information through meaningful visualizations. |
| PS-2                          | A researcher, data analyst, or policymaker studying the growth of electric vehicles in India.       | Analyze EV charging infrastructure, regional charging station distribution, vehicle range, efficiency, and market trends. | The datasets are large, unorganized, and difficult to interpret using raw tables alone.        | Traditional spreadsheets do not provide interactive insights or geographical visualizations for better understanding. | Unable to quickly identify trends and make informed decisions regarding EV adoption and infrastructure planning.      |